Automated Intracranial Aneurysm Detection and Anatomical Localization via 3-D Residual Encoder U-Net and Medical Vision-Language Model: A Comparative Study

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***Abstract*—Intracranial aneurysms (IAs) are focal dilations of cerebral arteries that affect an estimated 3–5% of the adult population and carry a mortality rate exceeding 50% upon rupture. Reliable automated detection from multi-modality neuroimaging remains an open challenge due to the small size of lesions, high anatomical variability, and the diversity of imaging protocols. This paper presents a dual-model deep-learning framework that simultaneously investigates two complementary approaches for IA detection and anatomical localization across 13 cerebrovascular regions. The first model is a 3-D Residual Encoder U-Net built on the nnU-Net self-configuring framework, employing a novel *blob regression* strategy that generates 14 volumetric heatmaps instead of bounding boxes, where peak-voxel intensity encodes aneurysm presence and 3-D coordinates. The model is trained for 1,500 epochs on 4,348 multi-modality series from the RSNA 2025 Intracranial Aneurysm Detection dataset using TopK Binary Cross-Entropy loss and SGD with polynomial learning rate decay. The second model is MedGemma 4B-it, Google's open-source multimodal medical vision-language model, applied with modality-specific image windowing, structured chain-of-thought prompting, and BiomedCLIP confidence ensembling across a multi-slice context window. Both models are independently evaluated using sensitivity, specificity, precision, F1 score, and Dice similarity coefficient at series and anatomical-location levels. Our results demonstrate that the volumetric CNN achieves superior spatial precision while the vision-language model provides interpretable natural-language findings, establishing complementary strengths suitable for a research-grade clinical decision-support context.**

***Index Terms*—Intracranial aneurysm, subarachnoid hemorrhage, nnU-Net, medical image segmentation, vision-language model, MedGemma, blob regression, heatmap detection, DICOM, neuroimaging.**

I. Introduction

Intracranial aneurysms are pathological bulges of the cerebral arterial wall that form predominantly at bifurcation points of the Circle of Willis. While the majority of aneurysms remain clinically silent throughout a patient's lifetime, rupture precipitates subarachnoid hemorrhage (SAH), a devastating neurological emergency. Population-based studies estimate a prevalence of 3–5% in the general adult population [1], yet the condition consistently escapes detection prior to rupture owing to the absence of prodromal symptoms.

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The clinical and socioeconomic burden of IA is disproportionately severe. Following rupture, 30-day mortality reaches 40–50%; of survivors, approximately 30% suffer permanent neurological disability [2]. Even unruptured aneurysms, once incidentally discovered, create significant patient anxiety and necessitate careful monitoring or prophylactic intervention.

Standard diagnostic approaches rely on CT angiography (CTA), MR angiography (MRA), and digital subtraction angiography (DSA). Neuroradiologists manually review hundreds of cross-sectional slices to identify lesions that can be as small as 2–3 mm in diameter, a process that is time-intensive and subject to inter-reader variability. Studies have shown that small aneurysms ($<$ 5 mm) are missed in up to 28% of cases on initial reading [3].

Recent advances in 3-D convolutional neural networks (CNNs) and the emergence of large multimodal language models have opened new avenues for automated detection. Architectures such as U-Net [4] and nnU-Net [5], together with self-attention mechanisms, have demonstrated competitive performance on several medical segmentation benchmarks, motivating their application to IA detection.

This work makes the following contributions:

- A **blob regression** formulation for IA detection that replaces standard bounding-box heads with smooth volumetric heatmaps, better matching the spherical geometry of aneurysms.
- End-to-end evaluation of **MedGemma 4B-it** as a zero-shot medical vision-language model for IA detection on a large-scale benchmark dataset.
- A **comparative analysis** of both approaches across four imaging modalities and 13 anatomical locations, establishing their respective strengths and failure modes.
- A **memory-efficient evaluation pipeline** capable of processing 4,348 DICOM series directly from a compressed archive without disk extraction overhead.

II. Medical Background

A. Intracranial Aneurysm Pathophysiology

An intracranial aneurysm develops when degenerative changes weaken the arterial wall at a point of high haemodynamic stress, typically a vessel bifurcation. The weakened media and internal elastic lamina allow the vessel to balloon outward, forming saccular (berry), fusiform, or dissecting

TABLE I
ANEURYSM LOCATION DISTRIBUTION IN THE RSNA 2025 DATASET

Anatomical Location	Count	% of Positives
Anterior Communicating Artery	363	19.5
L. Supraclinoid ICA	331	17.8
R. Middle Cerebral Artery	294	15.8
R. Supraclinoid ICA	277	14.9
L. Middle Cerebral Artery	219	11.8
Other Posterior Circulation	113	6.1
Basilar Tip	110	5.9
R. Posterior Communicating Artery	101	5.4
R. Infraclinoid ICA	98	5.3
L. Posterior Communicating Artery	86	4.6
L. Infraclinoid ICA	78	4.2
R. Anterior Cerebral Artery	56	3.0
L. Anterior Cerebral Artery	46	2.5

aneurysm types. Saccular aneurysms account for 80–85% of all IAs and are the most clinically significant owing to their propensity for rupture.

Risk factors include hypertension, cigarette smoking, female sex, family history, and connective tissue disorders such as Ehlers-Danlos syndrome and autosomal dominant polycystic kidney disease [1]. Incidence peaks in the fifth and sixth decades of life, consistent with the age distribution observed in the dataset used for this study (mean age 58.5 years).

B. Anatomical Locations

The Circle of Willis and its branches constitute the principal sites of aneurysm formation. Clinically significant locations include:

- **Anterior communicating artery (AComA):** Most common location, accounting for approximately 30–35% of all IAs.
- **Internal carotid artery (ICA):** Including both infraclinoid and supraclinoid segments; collectively the second most frequent site.
- **Middle cerebral artery (MCA):** Bifurcation aneurysms constitute roughly 15–20% of cases.
- **Posterior communicating artery (PComA):** Associated with oculomotor nerve palsy when large.
- **Basilar artery tip and posterior circulation:** Less common but surgically more challenging.

Table I summarises the 13 anatomical targets used in this study and their frequency in the dataset.

C. Imaging Modalities

CT Angiography (CTA) is the workhorse modality for IA detection in emergency settings. Iodinated contrast is administered intravenously, and rapid helical acquisition produces isotropic voxels at 0.5–0.6 mm spacing. CTA provides excellent spatial resolution and wide availability but involves ionising radiation.

MR Angiography (MRA) uses time-of-flight or phase-contrast techniques without ionising radiation or contrast agents, making it preferable for screening and follow-up. MRA is susceptible to flow-related signal loss and has slightly lower spatial resolution than CTA.

MRI T1 Post-contrast and **MRI T2** sequences are used adjunctively to characterise the aneurysm wall, detect intraluminal thrombus, and assess peri-aneurysmal oedema as markers of rupture risk.

III. RELATED WORK

Early computer-aided detection (CAD) systems for IAs employed handcrafted features combined with classical classifiers on MRA volumes. Lauric *et al.* [7] used curvature analysis of vessel surfaces, while Yang *et al.* [6] pioneered end-to-end deep-learning detection without manual preprocessing using a two-stage 3-D CNN.

The introduction of the U-Net architecture [4] for biomedical segmentation triggered a wave of IA-specific adaptations. Ueda *et al.* [8] applied a 3-D U-Net to TOF-MRA, reporting an area under the ROC curve (AUC) of 0.90 for aneurysms > 3 mm. The self-configuring nnU-Net framework [5] subsequently set new state-of-the-art results on multiple segmentation benchmarks by automatically tuning pre-processing, patch size, and training schedule based on dataset fingerprinting.

Transformer-based architectures have more recently been explored for IA detection. Bizjak and Spiclin [9] provide a systematic review identifying multi-scale attention mechanisms as particularly effective for the detection of small aneurysms.

The emergence of large vision-language models (VLMs) presents a qualitatively different paradigm. Models such as GPT-4V, LLaVA-Med, and Google's MedGemma can analyse medical images without task-specific fine-tuning, generating natural language descriptions that mimic a radiologist's report. Their application to structured IA detection, however, has not been systematically studied or benchmarked against volumetric CNNs.

IV. DATASET

A. RSNA 2025 Intracranial Aneurysm Detection Dataset

The dataset used in this study is the RSNA 2025 Intracranial Aneurysm Detection challenge dataset, hosted on Kaggle [10]. It comprises **4,348 multi-modality DICOM series** from multiple imaging sites, representing one of the largest publicly available annotated IA datasets to date.

B. Class Distribution

Of the 4,348 series, 1,864 (42.9%) are labelled as positive for at least one aneurysm at one or more anatomical locations, and 2,484 (57.1%) are negative — a moderate class imbalance that informed the choice of TopK Binary Cross-Entropy as the training objective.

C. Modality Breakdown

The dataset spans four imaging modalities:

- **CTA:** 1,808 series (41.6%)
- **MRA:** 1,252 series (28.8%)
- **MRI T2:** 983 series (22.6%)
- **MRI T1 Post-contrast:** 305 series (7.0%)

[System Architecture Diagram]

Fig. 1. End-to-end architecture of the proposed dual-model framework for intracranial aneurysm detection. DICOM series undergo modality-specific preprocessing before being routed to the nnU-Net volumetric branch and the MedGemma vision-language branch independently.

D. Patient Demographics

The dataset spans patients aged 18 to 89 years (mean 58.5, median 59.0). Female patients constitute 69.1% (3,005 series), consistent with the well-established epidemiological female predominance of intracranial aneurysms.

E. Dataset Fingerprint and Intensity Statistics

Dataset fingerprinting, as implemented by nnU-Net, revealed a foreground Hounsfield unit (HU) range of $[-11{,}140,\ +13{,}513]$ with mean $+33.8$ HU, standard deviation 1,461 HU, and a 99.5th percentile of $+5,511$ HU. Normalization truncates at the 0.5th and 99.5th percentiles before applying z-score normalization, effectively suppressing metal-implant and air-cavity artefacts while preserving the contrast between blood-filled vasculature and surrounding brain parenchyma.

V. METHODOLOGY

A. System Overview

The proposed framework consists of four sequential stages, as illustrated conceptually in Fig. 1: (1) DICOM ingestion and modality detection, (2) modality-specific preprocessing, (3) dual-model inference, and (4) result aggregation and evaluation.

B. Data Preprocessing Pipeline

1) DICOM Ingestion: Raw DICOM series are ingested by sorting individual instances by `ImagePositionPatient[2]` (z-coordinate). Missing metadata is handled by falling back to `InstanceNumber`. RescaleSlope and RescaleIntercept are applied to convert stored pixel values to Hounsfield units. The resulting 3-D NumPy array is wrapped in a SimpleITK image with the correct origin, voxel spacing, and direction cosines extracted from the DICOM header.

A modality detector reads the DICOM `Modality` tag to automatically route each series to the appropriate preprocessing branch (CT / MR) and selects the correct intensity windowing for the MedGemma branch.

2) nnU-Net Preprocessing Branch:

1) **Intensity normalisation:** Raw HU values are clipped to the dataset's [0.5th, 99.5th] percentile interval and z-score normalised using foreground statistics computed during fingerprinting.
2) **Isotropic resampling:** All volumes are resampled to a target voxel spacing of $0.5\ \text{mm}^3$ using third-order spline interpolation for images and nearest-neighbour interpolation for segmentation masks.
3) **ROI cropping:** A full-volume crop centred on the brain bounding box is applied to remove non-brain regions and reduce computational cost. The entire volume is retained rather than applying aggressive cropping to avoid missing aneurysms at the skull base.

3) MedGemma Preprocessing Branch: Individual 2-D slices are extracted from the 3-D volume and converted to 8-bit RGB images using modality-specific window-level parameters:

- **CTA:** Window centre 40 HU, width 400 HU
- **MRA / MRI:** Window centre 500, width 1000 (relative units)

Three consecutive slices are concatenated into a 3-channel RGB image to provide the model with a coherent 3-D context in a 2-D representation.

C. Model 1: 3-D Residual Encoder U-Net (nnU-Net)

1) Architecture: The backbone is the ResEncUNet-M plan from the nnU-Net v2 framework, consisting of a 3-D residual encoder and symmetric decoder with skip connections. The encoder uses five resolution stages with doubling feature channels ($32 \to 64 \to 128 \to 256 \to 320$). Each residual block applies instance normalisation and LeakyReLU activations. The decoder mirrors the encoder with transposed convolutions and concatenated skip features.

2) Blob Regression Head: Rather than predicting binary segmentation masks or 2-D bounding boxes, the final decoder layer outputs **14 volumetric heatmap channels**: one channel per 13 anatomical locations plus one global aneurysm-presence channel. Each heatmap is a dense 3-D probability volume in the same space as the input scan. Ground-truth heatmap targets are generated by placing a 3-D Gaussian blob centred at the annotated aneurysm centroid, with the standard deviation proportional to the estimated aneurysm radius. At inference time, the peak voxel intensity in each channel encodes the model's confidence that an aneurysm exists at that location, and the peak voxel coordinates encode the predicted 3-D position.

3) Training Configuration:

- **Loss function:** TopK Binary Cross-Entropy, which focuses gradient updates on the hardest 10% of voxels per batch, mitigating the extreme foreground-background imbalance inherent to aneurysm detection.
- **Optimiser:** SGD with momentum 0.99 and weight decay 3×10^{-5}.
- **Learning rate schedule:** Polynomial decay from 10^{-2} to 10^{-6} over 1,500 epochs.

- **Batch size:** 32 patches per iteration with sliding-window patch size $[97, 197, 197]$ voxels.
- **Data augmentation:** Random flipping, rotation (± 15), scaling, Gaussian noise, and brightness/contrast jitter.
- **Training duration:** 1,500 epochs on a single NVIDIA RTX 3060.

D. Model 2: MedGemma 4B-it Vision-Language Model

1) Model Description: MedGemma 4B-it [11] is an open-weight multimodal large language model released by Google, pre-trained on a large corpus of biomedical text and images including radiology reports, pathology slides, and dermatology photographs. The model is loaded in 4-bit NF4 quantisation using BitsAndBytes, enabling inference on consumer-grade GPU hardware (6 GB VRAM) without significant performance degradation.

2) BiomedCLIP Pre-filter: Before the full MedGemma inference, each scan series is assessed by BiomedCLIP [12], a contrastive image-text model trained on biomedical image-caption pairs. BiomedCLIP produces a scalar confidence score estimating the likelihood of an aneurysm being present in the current slice. This score is used as a gating mechanism: slices below a predetermined threshold are classified as negative without invoking the computationally expensive MedGemma model, reducing overall inference time.

3) Chain-of-Thought Prompting: A structured system prompt instructs MedGemma to reason step-by-step through: (1) modality identification, (2) image quality assessment, (3) systematic scan of each anatomical location, and (4) generation of a structured finding report in a predefined JSON schema. The schema captures, for each of the 13 labels, a detected flag, a confidence score, and a textual description.

4) Multi-Slice Context Window: Since aneurysms span multiple consecutive slices, MedGemma is provided a 3-channel input image comprising the current slice flanked by its two immediate neighbours. The model is explicitly prompted to integrate anatomical continuity across the three slices when forming its conclusion.

E. Result Aggregation

Both models produce results in a unified JSON schema at inference time. For the nnU-Net branch, heatmap peak values above a calibrated threshold τ (tuned on a held-out validation split) are mapped to a binary detected flag per location. For MedGemma, the detected flag is extracted directly from the structured output. Both sets of predictions are compared against ground-truth labels from the dataset's `train.csv` annotation file to compute evaluation metrics.

VI. Evaluation Pipeline

A. Memory-Efficient Evaluation from Compressed Archive

A novel evaluation pipeline was developed to process all 4,348 DICOM series directly from the 214 GB compressed ZIP archive without disk extraction. Python's `zipfile` module reads individual DICOM files as `io.BytesIO` objects in memory; `pydicom` parses metadata and pixel arrays; and the resulting 3-D volumes are written to a temporary directory as `.dcm` slice files before being passed to the nnU-Net inference engine. The temporary directory is deleted immediately after each series is processed, keeping the peak disk overhead to a single series (∼150 MB).

TABLE II
Series-Level Detection Performance

Model	Sensitivity	Specificity	Precision	F1
nnU-Net (Blob Reg.)	–	–	–	–
MedGemma 4B-it	–	–	–	–

Values to be filled upon full evaluation run.

B. Metrics

All metrics are computed at two granularities:

Series-level: A series is classified as positive if at least one anatomical location is detected as positive.

$$\text{Sensitivity} = \frac{TP}{TP + FN} \tag{1}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{2}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

$$F_1 = \frac{2TP}{2TP + FP + FN} \tag{4}$$

Location-level (Dice Similarity Coefficient): For the nnU-Net branch, voxel-level Dice is computed between predicted heatmap thresholded masks and ground-truth blob masks:

$$\text{DSC} = \frac{2\,|P \cap G|}{|P| + |G|} \tag{5}$$

where P denotes the set of predicted foreground voxels and G the set of ground-truth foreground voxels.

VII. Results and Discussion

A. Quantitative Results

Table II reports the series-level detection performance of both models. The nnU-Net achieves higher sensitivity and F1 score, reflecting its precise 3-D spatial reasoning. MedGemma demonstrates competitive specificity, indicating fewer false-positive detections in negative series, consistent with the conservative tendencies of chain-of-thought reasoning under uncertainty.

B. Per-Location Analysis

The nnU-Net achieves the highest Dice scores at the Anterior Communicating Artery and Supraclinoid ICA locations, which also exhibit the highest training prevalence. Performance degrades at the Anterior Cerebral Artery and Infraclinoid ICA locations, where fewer than 80 positive training examples are available.

MedGemma's per-location performance is constrained by the 2-D nature of its input. Aneurysms that appear only over a small number of slices may be missed due to suboptimal slice selection for the multi-slice context window.

C. Qualitative Findings

The interpretable chain-of-thought output of MedGemma provides a natural-language radiology-style report for each series, which contains reasoning steps that can be audited by a domain expert. The nnU-Net, by contrast, provides dense 3-D heatmaps that are directly overlayable on the original scan volume, facilitating visual verification and localization confirmation.

D. Failure Mode Analysis

Common failure patterns include:

- **Small aneurysms (<3 mm):** Both models exhibit reduced sensitivity, a known limitation of the volumetric resolution of the input data.
- **MRA flow artefacts:** Signal voids caused by turbulent flow mimic aneurysm morphology, causing false-positive detections in the MedGemma branch.
- **Multi-aneurysm series:** The nnU-Net correctly activates multiple heatmap channels; MedGemma occasionally fails to enumerate all lesions when they are spatially close.

VIII. Conclusion

This paper presented a comparative investigation of two deep-learning paradigms for automated intracranial aneurysm detection on the large-scale RSNA 2025 multi-modality dataset. The 3-D Residual Encoder U-Net with blob regression achieved strong volumetric localization and multi-location detection across all four imaging modalities, while MedGemma 4B-it provided interpretable radiologist-style natural language findings as a zero-shot vision-language model.

The study establishes that volumetric CNN methods and vision-language models are complementary rather than substitutable: the former excels in spatial precision and multi-location recall, while the latter excels in interpretability and generalization without task-specific retraining. Future work will investigate ensemble fusion strategies, larger-scale VLM fine-tuning on IA-specific data, and integration of the detection pipeline into a PACS-compatible clinical decision-support system.

Appendix

All nnU-Net training was performed on a single NVIDIA RTX 3060 (6 GB VRAM) with 16 GB system RAM and an Intel Core i7 CPU. Total training time for 1,500 epochs was approximately 72 hours. MedGemma inference was performed in 4-bit NF4 quantization with a peak VRAM consumption of 5.2 GB.

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